

readiness_preflight_20260826

T042 readiness preflight — will the 52-invariant gate pass?

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Tree state read (anchors, §11.4.6)

anchor	value
HEAD	a3e1141777914f3d71d62a6cd32a4c79c7f43842 (unchanged across the whole session)
branch	002-user-owned-downloads
untracked (-uall)	42 at start → 43 at end (concurrent writer added one)
modified	43
driver	scripts/pre_build_verification.sh, 1969 lines

scripts/commit-push-all.sh was **NOT** run. scripts/pre_build_verification.sh was **NOT** run in full. Every result below comes from executing an individual invariant or its delegated gate script in isolation, nice -n 19.

Invariant map

52 = 43 bracket-form [N/52] + 9 run_const_gate "N/52" invocations (33–38, 40, 42, 43). grep -c run_const_gate returns 12/13 — the extra hits are the function definition (L1325) and three comment lines (L1312, L1721, L1725). Union is 1..52 with no gap and no duplicate.

Control needles (§11.4.201(7)(b), same instrument + same path): [N/99]
→ 0 (instrument discriminates), invariant → 103 (instrument sees).

Blocking vs advisory

Exit logic is `FAIL_COUNT>0 → exit 1`. There is **no** coverage/PENDING layer. Only a `fail()` call blocks. Four invariants are structurally incapable of blocking because their failure branch is a bare echo "WARN: ...":

#	gate	why it cannot block
26	CM-BADGE-FRESHNESS-CHECK	echo " WARN: ..." on non-zero exit
39	CM-DANGEROUS-COMBINATION-FAIL-CLOSED	echo " WARN: ... (ADVISORY, non-blocking)"
40	CM-ORACLE-STRATEGY-NAMED-AND-INDEPENDENT	run_const_gate ... advisory
41	CM-OPENDESIGN-UI-SYSTEM	echo " WARN: ... (ADVISORY, non-blocking)"

Git index / status consumers

The driver barely uses git porcelain. `git status` appears only in **comments** explaining why it is *not* used (L1080-1082: it is structurally blind to the mtime-only defect class). The real enumerators are:

- **git ls-files** (tracked-only) — invariant 30's no-trace corpus, invariant 49.
- **find / shell glob** (filesystem, **sees untracked**) — invariant 16 (`find PROJECT_ROOT -maxdepth 1 -name '*.md' + find docs/ scripts/ -name '*.md'`), invariant 30 (`"$PROJECT_ROOT"/tests/{unit,pre_build,hooks}/test_*.sh`), and the delegated scanners.

Load-bearing ordering fact: `commit-push-all.sh` runs the long gate at **stage 3/6**, and `git add -A` only at **stage 5**. So at gate time every new file is still untracked — which means the `git ls-files` invariants cannot see them and only the `find/glob` invariants can. That is exactly where both new-file failures land.

Hand-maintained lists found

list	location	risk
DANGER_ROOTS (5)	driver L1511	BOB-205/214 — misses repo root, tools/, cmd/
exts (16, no sh)	cm_dangerous_combination_fail_closed.sh L457	BOB-213
BASH_TEST_SELF_RECURSIVE (2)	driver L1054	recursion guard
BASH_TEST_QUARANTINE (0)	driver L1064	ratchet, currently empty
inv-30 test glob (3 dirs)	driver L1196	omits tests /security/
.docs_chain/contexts (2)	.docs_chain/contexts/	narrow scope
excludes (9)	scanner L458	prune list

Verdict: the gate FAILS. Three blocking invariants, in driver order.

[16/52] CM-MARKDOWN-EXPORT-SYNC — 6 violations (reproduced 3×, identical)

Scope excludes docs/research/* and docs/qa/* but **not** docs/scripts/.

- docs/scripts/test_gitignore_swallow_is_loud.{html,pdf}
MISSING — caused directly by the new untracked docs/scripts/test_gitignore_swallow_is_loud.md. This is the §11.4.18 companion doc written for the new test script; authoring it is what created the invariant-16 violation.
- docs/Issues.{html,pdf} + docs/Fixed.{html,pdf} **STALE** — .md mtime 1787774388 vs exports 1787772253–1787772292 (~35 min newer). **MOVING TARGET**: these six files are all M and owned by a concurrent writer.

Remediation the driver implies: bash scripts/
generate_markdown_exports.sh.

[30/52] CM-BASH-UNIT-TESTS-EXECUTED — a new RED test fails the gate

Enumerates by filesystem glob → runs the new untracked suites.
Executed exactly as the driver does (BOBA_PREBUILD_NESTED=1 timeout 300 bash <suite>):

suite	exit
tests/pre_build/ test_bob205_danger_roots_scope.sh (NEW)	1 — RED
tests/pre_build/ test_check_cm_lan_routes_authenticated.sh (NEW)	0 (115/0, 56s)
tests/unit/ test_ddos_sibling_retry_after_classification.sh (NEW)	0 (32/0)

BASH_TEST_FAILED>0 → fail(). The BOB-205 suite is a RED test for an unfixed defect — it is supposed to fail, and invariant 30 has no RED-test allowance.

[48/52] CM-RUNTIME-DEPS-PARITY — 8 divergences (pre-existing, state-dependent)

venv 3.4.9/3.32.2/3.18/0.27.3/1.4.1/0.4.2/0.52.1 vs container counterparts, plus CPython 3.14.6 (venv) vs 3.12.13 (container). Tracked as BOB-154, blocked on BOB-158. Not caused by any new file. It compares the live .venv against the running qbittorrent-proxy container, so it is state-dependent: with the container down the gate SKIPS and this failure disappears.

Per-new-file prediction

file	invariants that see it	prediction
docs/scripts/test_gitignore_swallow_is_loud.md	16 (find docs/)	FAIL — no .html/.pdf twins
docs/qa/{BOB-186,191,196,198,205,206,212,T028-uid100999,T041}/**	none	PASS — inv 16 excludes docs/qa/*; inv 19/20 read only docs/

file	invariants that see it	prediction
		QA_DISCOVERY_LEDGER. never the filesystem
tests/pre_build/ test_bob205_danger_roots_scope.sh	30 (glob, executes)	FAIL — RED by design
tests/pre_build/ test_check_cm_lan_routes_authenticated.sh	30	PASS (115/0)
tests/unit/ test_ddos_sibling_retry_after_classification.sh	30	PASS (32/0)
tests/security/ test_gitignore_swallow_is_loud.sh	none	PASS by invisibility — orphan guard
scripts/pre_build/ check_cm_lan_routes_authenticated.sh	none (not wired into the 52)	PASS
scripts/pre_build/lib/cm_export_charset_scan.py	23 (PASS), 39 (adds 1 WARN hit)	PASS
scripts/pre_build/ cm_export_charset_valid.baseline	50	PASS (343 scanned, 0 missing, at baseline)
scripts/pre_build/ tighten_cm_export_charset_baseline.sh	none	PASS
config/lan_route_auth_policy.yaml	none	PASS
qBitTorrent-go/internal/jackettapi/ bob204_failclosed_test.go	39 only (advisory)	PASS
specs/**/evidence/*.md (11 new)	none — inv 16 scans only root, docs/ scripts/	PASS

§11.4.18 script-documentation is NOT gated. grep -E

'11\4\18[^0-9]' over the driver returns **zero** (the earlier count of 1 was 11.4.18 prefix-matching 11.4.180+). There is no CM-SCRIPT-DOCS-SYNC among the 52. New scripts without a companion doc are therefore not a gate failure — the irony is that *writing* one (docs/scripts/test_gitignore_swallow_is_loud.md) is what tripped invariant 16.

§11.4.44 revision headers are NOT gated either: `grep -cE 'REVISION-HEADER|revision_header|Revision:\*\*' → 0` (control needle CM-BADGE-FRESHNESS → 3). New .md without headers will not fail.

Residue scan (inv 23): 273 files, 0 hits, 1 audited waiver — PASS. The count already sits at the predicted 273; no new file trips it.

§11.4.224 test-first: no invariant demands a paired mutation or fixture for a new gate script. Invariant 30 only demands that whatever suites *exist* pass.

Task 3 — the three-hole interaction

Answer: no. Invariant 39 does not pass today, and fixing the holes cannot make T042 fail.

Two independent reasons, both measured:

1. **Structural.** Invariant 39’s non-zero branch is `echo " WARN: ... (ADVISORY, non-blocking per §11.4.234 ...)"`. It never calls `fail()`. `FAIL_COUNT` is untouched regardless of hit count.
2. **Empirical.** Invariant 39 is *already* WARNING. Running its own gate over its own `DANGER_ROOTS` at this tree state:

root	hits
download-proxy/src	28
plugins	50
scripts	2
frontend/src	0
qBitTorrent-go	not scanned (stay-off); per BOB-191 structurally unanalysable → ~0
total	~80

So the premise “invariant 39 currently PASSes because of the holes” is false — it currently WARNs with ~80 hits and contributes nothing to the exit code either way.

Hole probes (a hit is a LEAD, §11.4.194(6)(b) — all triaged to the line):

probe	result	triage
	1 hit	

probe	result	triage
scripts ext=sh (BOB-213)		scripts/codegraph_validate.sh:81 — a real try{...}catch(e){} inside an embedded Node -e one-liner. Genuine pattern match, but guarded downstream by n=0 default + ["\$MCP_NODES" -gt 0] fail-closed assertion. Low severity.
tools default ext (BOB-214)	3 hits	tools/ plugin_update_automation.py:189,198,215 — silent default returns in first-party Python. Real and genuinely unscanned.
cmd default ext	0 hits	the RED test's cmd/boba-ctl needle is synthetic; no live hit
config default ext	43 hits	FALSE POSITIVE class — all in config/qBittorrent/nova3/engines/*.py, verified git check-ignore → GITIGNORED. These are the <i>installed copies</i> of plugins/ (already a DANGER_ROOT), i.e. duplicates of hits already counted. config/ is correctly excluded.
challenges ext=sh	0 hits	clean

Net: fixing all three holes moves the advisory WARN from ~80 to ~84. **Zero effect on T042's exit code.** T042's outcome is not an artifact of scanner blindness.

Verified-PASS roster

Executed in isolation, all exit 0 / fail()-free: **1–15** (extracted, 0 fails), **17** (validate 218 items OK; diff in-sync 218/218; DB md5 stable across the read), **18, 19–20** (extracted, ledger 28==28), **21–22** (extracted), **23** (273 files, 0 hits), **24** (docs_chain 2/2 in-sync), **25** (5 signatures below threshold), **27** (170 files), **28** (324), **29** (324), **31–32** (17/17 lockstep), **33–38** (const-gates), **42–43** (honest SKIP, wired blocking), **44** (5 services), **45, 46** (8 counts), **47** (2 builder stages), **49, 50** (343 exports, baseline 0), **51** (92 inputs), **52** (79 inputs).

Honest boundary (§11.4.6)

1. **Invariant 30 is only 3/38 verified.** It runs 40 globbed suites minus 2 structurally excluded. I ran the **3 new** ones. The other **35 are unverified** — any one failing adds to the same single `fail()`. This is the largest residual uncertainty and it can only widen the failure, never narrow it.
2. **qBittorrent-go was not scanned** (stay-off constraint). Its inv-39 contribution is assumed ~0 on BOB-191's structural-unanalysability finding, not measured here.
3. **Invariant 48 is state-dependent, not tree-dependent** — its verdict flips with whether the qbittorrent-proxy container is running.
4. **Four of the six invariant-16 violations are on concurrently-owned files** (`docs/{Issues,Fixed}. {html,pdf,md}`). Another agent may regenerate them at any moment. Only the two `docs/scripts/test_gitignore_swallow_is_loud.*` misses are attributable to this session's new files and stable.
5. **Cross-invariant interactions were not simulated.** Invariant 30's no-trace corpus check fails if any suite leaves an mtime trace on a tracked file; I ran suites individually, not under that guard, so a trace-induced inv-30 failure would not have shown up in my runs.
6. **Whole-driver-only effects unobservable:** ordering, the `BOBA_PREBUILD_NESTED` sentinel path, total wall-clock, and any shared state between invariants.

New finding surfaced by this preflight

`tests/security/test_gitignore_swallow_is_loud.sh` (a BOB-212 RED test authored this session) sits in a directory **no invariant executes** — inv 30's glob covers only `tests/unit`, `tests/pre_build`, `tests/hooks`. This is a recurrence of the exact orphan-guard class the driver's own comment at L1190-1195 records as having already happened twice ("the SAME orphan-guard class that stranded `tests/pre_build/` recurred one directory over"). Third occurrence, one directory over again.